

Sumin Hong

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INTERESTS

Human-Computer Interaction, Human-AI Interaction, Multimodal Fusion, Behavior Modeling, Visual Generation, Applied Machine Learning

EDUCATION

University of Notre Dame

Ph.D, Computer Science and Engineering
Advisor: Dr. Toby Jia-Jun Li

Notre Dame, IN

Aug. 2024 – Present

Seoul National University of Science and Technology (SeoulTech)

Seoul, KR

Northumbria University (*dual degree)

London, UK

B.S, Information Technology Management

Mar. 2017 – Feb. 2024

Honors: Summa Cum Laude, First-class

PUBLICATIONS

C5. **MORA: AI-Mediated Story-Based practice for Speech Sound Disorder from Clinic to Home**

(Under Review) IUI 2026: ACM Conference on Intelligent User Interface

Sumin Hong, Xavier Briggs, Qingxiao Zheng, Yao Du, Jinjun Xiong, Toby Jia-Jun Li

C4. **Vistoria: A Multimodal System to Support Fictional Story Writing through Instrumental Text-Image Co-Editing**

(Under Review) CHI 2026: ACM Conference on Human Factors in Computing Systems

Kexue Fu, Jingfei Huang, Long Ling, Sumin Hong, Yihang Zuo, Ray LC, Toby Jia-Jun Li

C3. **CreativeConnect: Supporting Reference Recombination for Graphic Design Ideation with Generative AI**

CHI 2024: ACM Conference on Human Factors in Computing Systems

DaEun Choi, Sumin Hong, JeongEun Park, John Joon Young Chung, Juho Kim

C2. **Class-wise Buffer Management for Incremental Object Detection: Effective Buffer Training Strategy**

ICASSP 2024: IEEE International Conference on Acoustics, Speech and Signal Processing

Junsu Kim, Sumin Hong, Chanwoo Kim, Jihyeon Kim, Yihalem Yimolal Tiruneh, Jeongwan On, Jihyun Song, Sunhwa Choi, Seungryul Baek

J1. **Automatic Examination of Condition of Used Books with YOLO-based Object Detection Framework**

Computer Systems Science and Engineering, 47(2), 1611-1632, 2023 (JIF=4.397)

Sumin Hong, Jin-woo Jeong

C1. **Ensemble of Multi-task Learning Networks for Facial Expression Recognition In-the-Wild with Learning from Synthetic Data**

ECCV 2023 4th Workshop on Affective Behavior Analysis in-the-Wild: IEEE/CVF European Conference on Computer Vision

Jae-yeop Jeong¹, Young-gi Hong¹, Sumin Hong¹, Jiyeon Oh¹, Jin-woo Jeong

POSTERS, DEMOS, WORKSHOPS

P3. **Proactive Support through floating windows in Information Foraging with User Attention Modeling via Eye-tracking**

CHI 2025 Computational UI Workshop

Sumin Hong, Brandon Foresth, Toby Li

P2. An Effective Buffer Training Strategy for Class-Incremental Object Detection

CVPR 2023 Workshop on Continual Learning in Computer Vision

Junsu Kim Sumin Hong, Chanwoo Kim, Hoseong Cho, Yihalem Yimolal Tiruneh, Jihyeon Kim, Eunseo Kim, Jihyun Song, Sunhwa Choi, Seungryul Baek

P1. Visualizing Instructor's Gaze Information for Online Video-based Learning: Preliminary Study

ETRA 2022 Workshop on Eye Tracking in Learning and Education

Daun Kim¹, Jae-yeop Jeong¹, Sumin Hong, Namsub Kim, Jin-woo Jeong

EXPERIENCE

Applied ML/AI Engineer

Lucid Dream

Seoul, KR

Nov. 2023 – Feb. 2024

• Led the development of a skeleton-based anomaly detection model for children's rehabilitation monitoring

Research Intern

KAIST Interaction Lab (KIXLAB)

Daejeon, KR

Jun. 2023 – Feb. 2024

Mentor: Juho Kim, DaEun Choi

[C3]

Research Intern

UNIST Vision and Learning Lab (UVLL)

Ulsan, KR

Jan. 2023 – Jun. 2023

Mentor: Seungryul Baek, Junsu Kim

[C2, P2]

Research Intern

SeoulTech Interaction Lab (IXLAB)

Seoul, KR

Jun. 2021 – Dec. 2022

Mentor: Jin-woo Jeong, Jae-yeop Jeong

[C1, J1, P1]

HONORS AND AWARDS

Faculty Choice Research Award Poster Competition | University of Notre Dame

Mar. 2025

Graduate Best Award SeoulTech

Dec. 2023

Full Tuition Scholarship Merit-based | SeoulTech

2017, 2023

SeAH Haiam Scholarship Merit-based | SeAH Corporate

2023

3rd place 4th Workshop on Affective Behavior Analysis in-the-wild | ECCV 2022

Jul. 2022

Outstanding Paper Award Winter Annual Conference of KICS

Jan. 2022

Moksan Scholarship Merit-based | Daewon Kangup Co.Ltd.

2020

TEACHING

Teaching Assistant CSE20110: Discrete Mathematics | University of Notre Dame

Fall 2024

Teaching Assistant ITM146091: Programming Language | SeoulTech

Fall 2020

ACADEMIC SERVICE

Reviewer CHI 26, CHI LBW 25